

LINJIE LYU

July 05, 1997

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🌐 <https://linjielyu.github.io/>

RESEARCH INTERESTS

- Diffusion Models Based 2D/3D Generation and Editing
- Global-illumination-aware Inverse Rendering, 3D Reconstruction, Relighting
- Radiance Fields, Efficient 3D Gaussian Splatting
- Uncertainty Quantification

EDUCATION

Max-Planck-Institut für Informatik, Saarbrücken, Germany
Ph.D. Student in Computer Vision and Computer Graphics

June 2021 — January 2026

Universität des Saarlandes, Saarbrücken, Germany
Graduate School in Computer Science

October 2019 — March 2021

Tsinghua University, Beijing, China
Bachelor in Physics and Mathematics

September 2015 — July 2019

* College Scholars; University Honors

Sichuan Mianyang High School, Mianyang, China
* Silver Medal, Chinese Physics Olympiad

September 2012 — June 2015

PUBLICATIONS

- **L. Lyu**, A. Tewari, J. Chen, T. Leimkühler, C. Theobalt. *Faster 3D Gaussian Splatting Convergence via Structure-Aware Densification*. Conference Proceedings, SIGGRAPH 2026
- X. Zhou, J. Chen, P. Rao, T. Teufel, **L. Lyu**, T. Minasian, O. Sotnychenko, X.-X. Long, M. Habermann, C. Theobalt. *OLATverse: A Large-scale Real-world Object Dataset with Precise Lighting Control*. CVPR 2026 (Oral).
- **L. Lyu**, V. Deschaintre, Y. Hold-Geoffroy, M. Hasan, J. Yoon, T. Leimkühler, C. Theobalt, I. Georgiev. *IntrinsicEdit: Precise Generative Image Manipulation in Intrinsic Space*. ACM Transactions on Graphics, SIGGRAPH 2025.
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Manifold Sampling for Differentiable Uncertainty in Radiance Fields*. Conference Proceedings, SIGGRAPH Asia 2024
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Diffusion Posterior Illumination for Ambiguity-aware Inverse Rendering*. ACM Transactions on Graphics, SIGGRAPH Asia 2023.
- **L. Lyu**, A. Tewari, T. Leimkühler, M. Habermann, C. Theobalt. *Neural Radiance Transfer Fields for Relightable Novel-view Synthesis with Global Illumination*. ECCV 2022 (Oral).
- **L. Lyu**, M. Habermann, L. Liu, M. B. R., A. Tewari, C. Theobalt. *Efficient and Differentiable Shadow Computation for Inverse Problems*. ICCV 2021.

INTERSHIP

Adobe, London, UK

June 2024 — September 2024

Developed diffusion model-based and global illumination-aware image editing techniques utilizing intrinsic decomposition and identity preservation for tasks such as relighting, material editing, and object insertion/removal on real 2D images.

SERVICES

Reviewer

SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, 3DV

Teaching

Computer Vision and Machine Learning for Computer Graphics